

Thirulok Sundar Mohan Rasu

Ann Arbor, MI 48105 | +1 (248) 704-2911 | thirulok@umich.edu | thiruloksundar.github.io | github.com/Thiruloksundar |
linkedin.com/in/thirulok-sundar-mohanrasu

Education

University of Michigan — Ann Arbor

Aug 2025 – Dec 2026

M.S., Electrical and Computer Engineering

GPA: 4.0 / 4.0

Relevant courses: *Algorithmic Robotics, Mobile Robotics, Matrix Methods for ML & Signal Processing*

Indian Institute of Technology (ISM) Dhanbad

Dec 2021 – May 2025

B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)*

GPA: 8.30 / 10

Skills

Languages: Python, C++, MATLAB

3D Vision & Reconstruction: 3D Gaussian Splatting (3DGS), Multi-view 3D/4D Reconstruction, Depth Estimation, SLAM, LiDAR, Sensor Fusion

Physical-AI & Robotics: ROS2, Nav2, State Estimation, Robot Perception, Object Detection & Segmentation, VLMs, Embodied-AI Datasets (DROID)

ML & Tools: PyTorch, OpenCV, HuggingFace, TensorFlow, GitHub, Linux

Research Experience

University of Michigan (Prof. Jeong Joon Park)

Aug 2025 – Dec 2025

Graduate Research Assistant

- Applied **3D Gaussian Splatting (3DGS)**, **SpatialTracker**, and **VGGT** to the **DROID embodied-AI manipulation dataset** to build a **4D scene reconstruction pipeline**; generates dynamic 3D scene representations from robot manipulation video sequences for physical-AI perception research
- Designed **data collection pipelines and evaluation protocols** to assess reconstruction quality, temporal consistency, and coverage across diverse manipulation scenes and viewpoint conditions

HDR Lab — University of Michigan

Jan 2026 – Present

Graduate Research Assistant

- Developing real-time **3D depth estimation** algorithms for visuo-tactile sensors on robotic arms; built **iOS app** for automated physical-AI data collection enabling large-scale dataset generation for model training

University of Central Florida — CRCV Lab (Prof. Yogesh Rawat)

Feb 2024 – Dec 2024

Research Intern

- Designed **evaluation harnesses** for **Video-LLMs**; defined fine-grained metrics for actor/action/object/environment recognition and targeted question types to systematically **surface model failures and hallucinations**
- Built annotated **video-activity dataset** with structured evaluation protocol; defined **coverage and diversity metrics** across scenarios to benchmark model robustness

Mowito

Mar 2023 – Jul 2023

Robotics Software Intern

- Built modular **robot perception pipeline** using **Multi-head Mask R-CNN** for instance segmentation and object detection; integrated into production warehouse automation pipelines
- Implemented **perceptual hashing** for large-scale dataset deduplication (30% size reduction); developed order-fulfillment optimization using **Leiden clustering** for crate grouping

Selected Projects

Semantic LiDAR-SLAM with Language-Guided Navigation (GitHub)

2026

Integrated **Cartographer SLAM** with **Grounding DINO** in ROS2; SLAM builds a metric map while DINO populates a persistent **semantic object map** — natural-language commands trigger **Nav2** navigation goals, enabling **zero-shot object-goal navigation** without environment-specific training.

VLA Model Fine-tuning for Robot Manipulation (GitHub)

2025

Fine-tuned OpenVLA (7B vision-language-action model) on the **LIBERO** embodied-AI manipulation benchmarks via **LoRA**; deployed on UMich HPC (NVIDIA A40). Training loss: 19.08 → 1.16 over 3 epochs.

Multi-view 3D Scene Reconstruction Analysis

2025

Systematically analyzed **MASt3R** failure cases under challenging capture conditions (occlusion, viewpoint variation, lighting); collected custom scene dataset and documented limitations — directly relevant to **3DGS-based simulation** robustness.